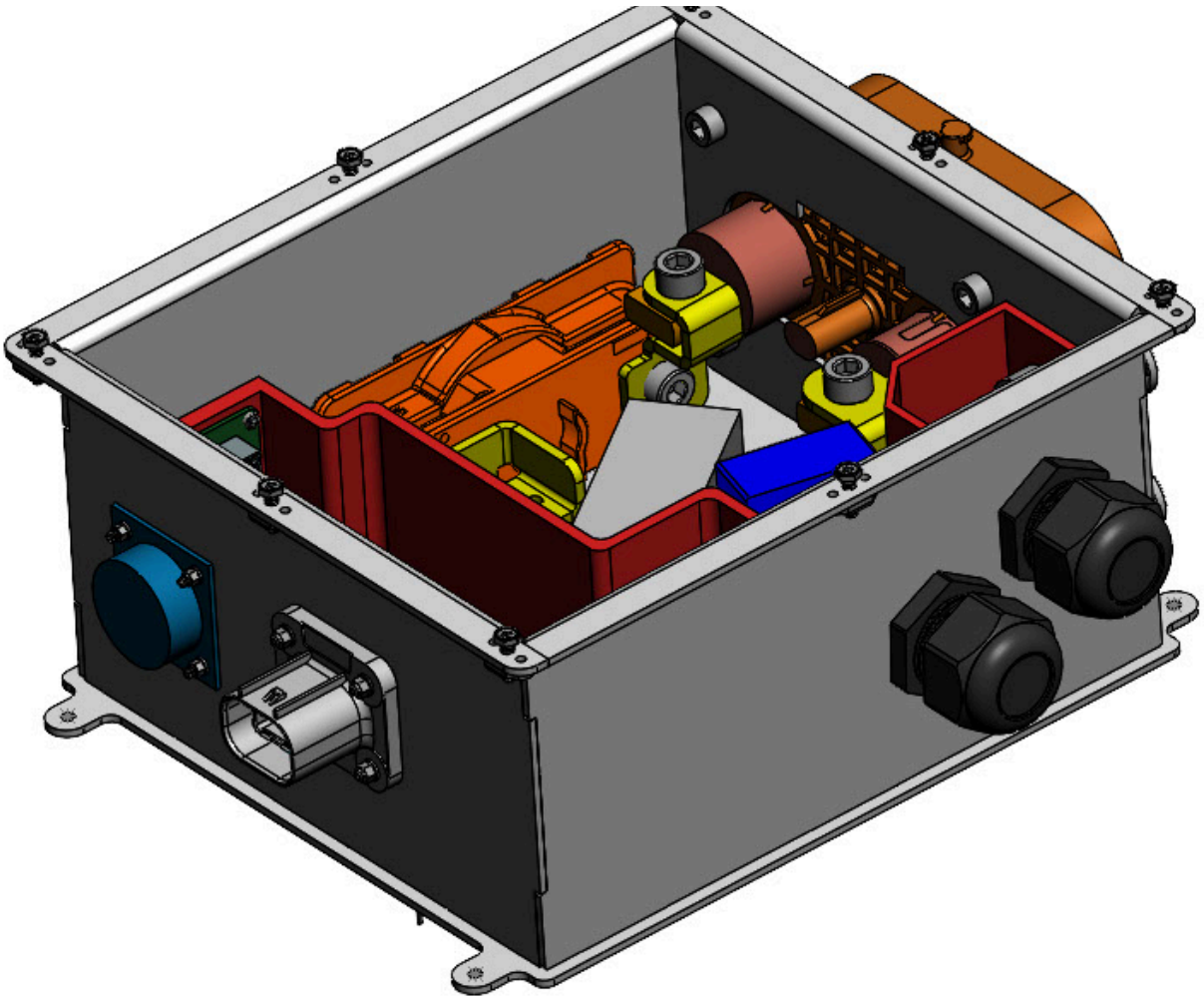


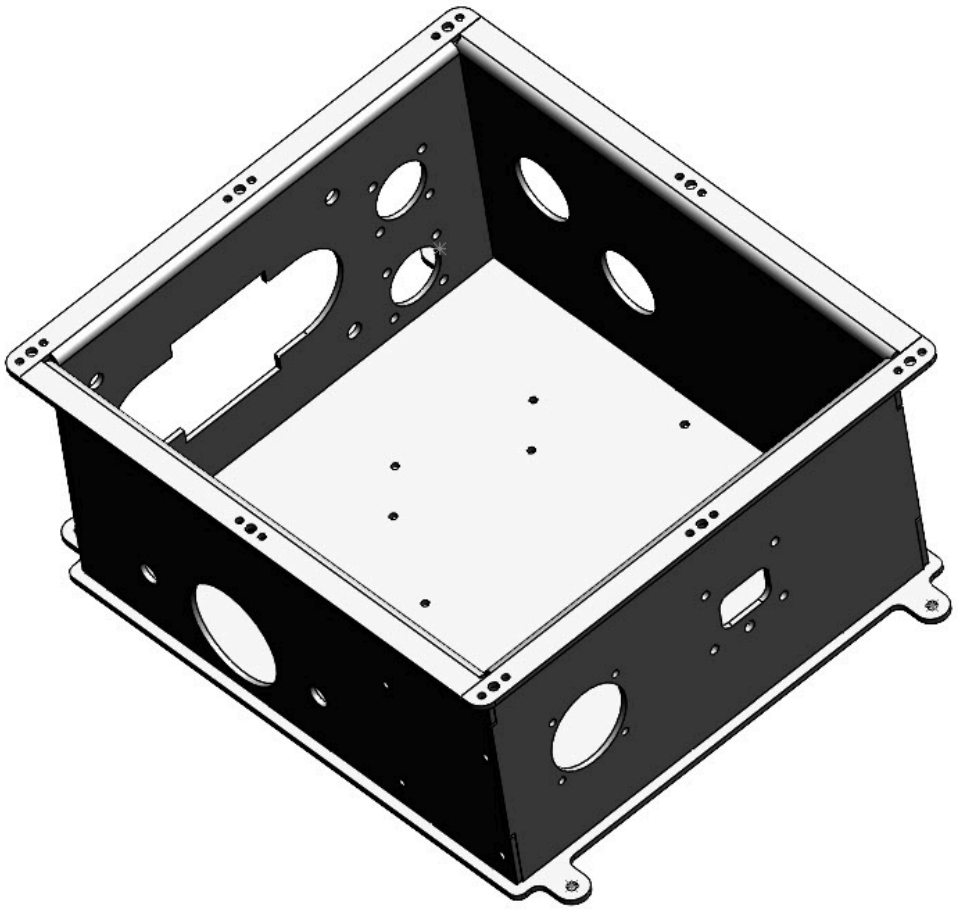
MY JOURNEY

By Jesus Estrada

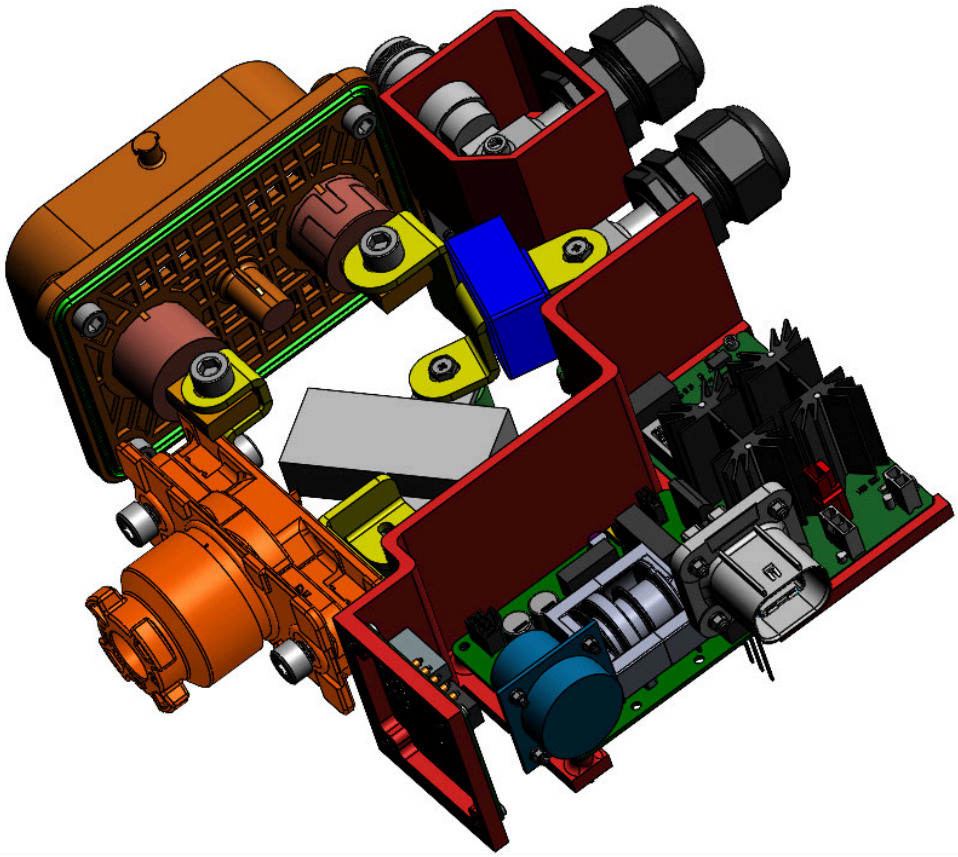
HIGH VOLTAGE ELECTRONICS HOUSING



Took initiative to design and build a sheet metal housing for the team's high-voltage electronics package after the original designer shifted focus to higher-priority tasks. Using SolidWorks Sheet Metal, I developed the enclosure with 0.090-inch Aluminum 5052, selected for its form ability and lightweight properties compared with steel. I chose Aluminum 5052 over 6061 because it is better suited for bending and forming, while 6061 is stronger but more susceptible to cracking during fabrication. Since the housing primarily served as an enclosure rather than a structural component, extensive structural analysis was not required. The completed housing satisfied the project requirements and provided secure protection for the electronics.



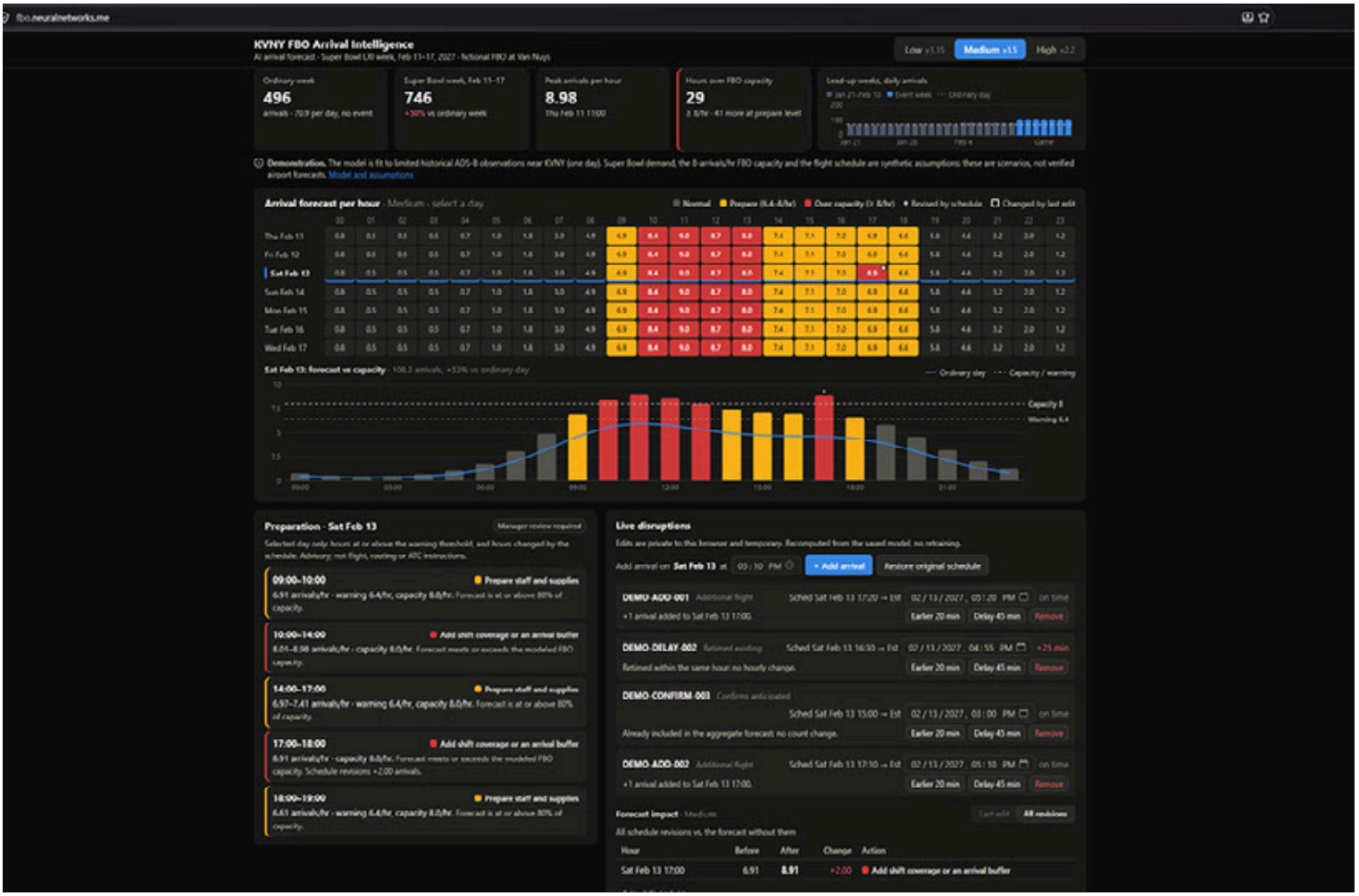
Housing I made



Packaged Electronics

FIRST PLACE IN HACKATHON

For the NCBAA Hackathon, our team developed an AI-powered platform focused on improving aviation operations, safety, sustainability, and community impact. We trained the AI using aviation research, industry articles, and relevant data, then created a website that made the information accessible through an interactive user experience. I contributed to researching and organizing the training material, developing the project concept, refining the presentation, and helping communicate how the solution could support smarter decision-making across the aviation industry. Our project earned first place in the NCBAA track and was presented to judges and representatives from companies including Google, Adobe, and IBM.

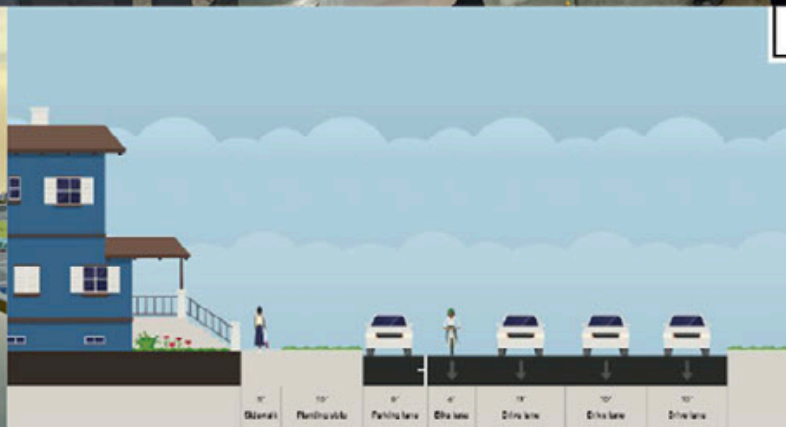


ENGINEERING TEAM LEADER AT LOS ANGELES METRO









As a Student Intern Team Leader, I helped lead a multidisciplinary team in developing a Complete Streets proposal for Los Angeles's Vermont Avenue corridor. I coordinated team responsibilities, guided research and design discussions, and used 3DStreet and SolidWorks CAD to visualize roadway, pedestrian, transit, and landscaping improvements. Our proposal focused on safer bike lanes, wider sidewalks, dedicated bus lanes, improved transit stops, native plants, and better access near the Vermont/Athens C-Line Station.

